

Thomas Westhoefer

Senior Data Platform Engineer

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Senior Data Engineer and Data Platform Architect with 10+ years of experience building **scalable Lakehouse architectures** across **AWS** and **Azure** using **Databricks**, **Snowflake**, **BigQuery**, **Delta Lake**, and **Apache Spark**. Proven leader in designing **distributed ETL/ELT systems** with **Infrastructure-as-Code** (Terraform), **Kubernetes** orchestration, CI/CD automation, and implementing data modeling, **KPI governance**, **A/B experimentation**, advanced SQL analytics, **ML feature pipelines**, and robust data quality and observability frameworks to ensure reliability, performance, and cost optimization at enterprise scale.

Core Technologies

- ▮ **Programming Languages:** Python, SQL, PySpark
- ▮ **Data Platforms & Lakehouse:** Snowflake, Google BigQuery, Databricks, Delta Lake, Lakehouse architecture (bronze/silver/gold), Dimensional modeling, Star schema, Semantic layer design
- ▮ **Data Engineering & Processing:** ETL/ELT pipelines, Apache Spark, dbt, Apache Airflow, Incremental pipelines, CDC processing, Batch processing, Streaming pipelines
- ▮ **Data Warehousing & Modeling:** Snowflake, Google BigQuery, Delta Lake, Amazon Redshift, Lakehouse architecture (bronze/silver/gold), Dimensional modeling, Star schema, Semantic layer design
- ▮ **Streaming & Event Systems:** Apache Kafka, AWS Kinesis, Google Pub/Sub
- ▮ **Cloud & Infrastructure:** Microsoft Azure, Amazon Web Services (S3, EC2), Terraform, Infrastructure-as-Code (IaC), Kubernetes, Docker
- ▮ **Azure Data Services:** Azure Databricks, Azure Data Lake Storage (ADLS), Azure Data Factory (ADF), Azure Synapse Analytics
- ▮ **DevOps & CI/CD:** Git, GitHub Actions, GitLab CI, Automated testing & deployment pipelines
- ▮ **Data Quality & Observability:** Unit testing, Integration testing, dbt tests, Great Expectations, Data quality frameworks, Freshness SLAs, Monitoring & alerting, Data observability tools (Monte Carlo concepts), Schema validation, Role-Based Access Control (RBAC)
- ▮ **Analytics & Business Intelligence:** Advanced SQL storytelling, A/B testing, KPI ownership, Cohort analysis, Metric governance, Looker, Tableau, Power BI
- ▮ **Machine Learning Data Enablement:** Feature engineering pipelines, Training datasets, Inference datasets, Real-time feature freshness systems

Professional Experiences

Senior Data Engineer

Sift – San Francisco, CA, USA

April 2021 – Present

- ▮ Architected and evolved a cloud-native Lakehouse data platform across **AWS** and **Azure** environments, leveraging **Snowflake**, **BigQuery**, **Azure Databricks**, and **Delta Lake** to process **billions of monthly events** supporting fraud detection, experimentation, analytics, and ML inference systems.

- ▮ Designed and implemented scalable **ETL/ELT pipelines** using **Python**, **PySpark**, **Apache Spark**, **dbt**, and **Apache Airflow**, running distributed workloads on **Databricks (AWS & Azure)**, delivering curated analytics-ready and ML-ready datasets.
- ▮ Built event-driven ingestion systems leveraging **Apache Kafka**, **AWS Kinesis**, and **Google Pub/Sub**, landing raw data into **S3** and **Azure Data Lake Storage (ADLS)** for downstream Lakehouse processing.
- ▮ Implemented **bronze/silver/gold** data layers using **Delta Lake** on **Databricks**, enabling **ACID-compliant** distributed transformations and optimized large-scale analytical processing.
- ▮ Provisioned and managed cloud-based data infrastructure using **Terraform** and **Infrastructure-as-Code (IaC)** principles, ensuring reproducible and version-controlled environment deployments.
- ▮ Containerized Spark-based workloads using **Docker** and orchestrated runtime environments with **Kubernetes**, supporting scalable distributed execution across cloud environments.
- ▮ Designed dimensional and ML feature schemas within **Snowflake**, **BigQuery**, and **Azure Databricks**, standardizing datasets consumed by **100+ dashboards** and ML pipelines.
- ▮ Integrated **Azure Data Factory (ADF)** and **Airflow** for orchestrating cross-cloud data workflows, improving reliability and pipeline observability.
- ▮ Owned **KPI definitions** and metric governance across fraud, experimentation, and product domains, ensuring consistency across **Looker**, **Tableau**, and **Power BI** semantic layers.
- ▮ Supported **A/B testing frameworks** by delivering validated experiment datasets and applying advanced **SQL storytelling** techniques (window functions, cohort segmentation, statistical comparisons) to evaluate experiment outcomes.
- ▮ Implemented comprehensive data quality frameworks including **dbt tests**, **schema validation**, **referential integrity checks**, **freshness SLAs**, and **Great Expectations-style validations** to reduce data regressions by **40–50%**.
- ▮ Established **unit testing** and integration testing practices for transformation logic and pipeline orchestration workflows, integrating automated validation into **CI/CD pipelines**.
- ▮ Integrated observability and monitoring solutions including **logging**, **alerting**, and **anomaly detection** (Monte Carlo-style monitoring concepts) to proactively identify pipeline failures and data anomalies, reducing incidents by **30–40%**.
- ▮ Built and maintained **CI/CD pipelines** using **GitHub Actions** and **GitLab CI** to automate testing, deployment, and controlled releases of **dbt models** and **Spark workloads**.
- ▮ Led architectural design reviews, mentored engineers, partnered with product and ML teams to translate ambiguous business problems into scalable data solutions, and optimized **Snowflake** and **Spark workloads** achieving **15–25% cost savings**.

Environment: AWS, Microsoft Azure, Azure Databricks, Azure Data Lake Storage (ADLS), Azure Data Factory, Snowflake, BigQuery, Apache Spark, Delta Lake, Kafka, Kinesis, Pub/Sub, dbt, Airflow, Terraform, Kubernetes, Docker, GitHub Actions, Looker, Tableau, Power BI, Python, SQL

Senior Data Engineer

WorkFusion – New York, NY, USA

January 2016 – March 2021

- ▮ Designed and implemented scalable **ETL/ELT pipelines** using **Python** and **SQL** to process high-volume operational and behavioral data across multiple enterprise client environments.
- ▮ Built reusable dimensional data models (star schema, fact/dimension tables, semantic layers) to standardize KPI tracking and reporting consistency across business domains.
- ▮ Developed advanced **SQL** transformations leveraging window functions, aggregations, cohort analysis, and performance optimization to enable executive-level analytics and storytelling.
- ▮ Delivered curated and governed datasets powering **Tableau** and **Power BI** dashboards used for operational reporting, executive summaries, and client-facing analytics deliverables.

- ▮ Owned KPI definitions and reporting logic across enterprise engagements, reducing metric inconsistencies and improving stakeholder trust in analytical outputs.
- ▮ Implemented data validation and reconciliation frameworks including schema validation, data consistency checks, and integration-level verification across reporting pipelines.
- ▮ Collaborated cross-functionally with product managers, analysts, and enterprise stakeholders to translate business requirements into scalable data models and analytics workflows.
- ▮ Managed multiple concurrent data environments, maintaining SLA adherence, optimizing query performance, and ensuring production stability across enterprise client engagements.

Environment: Python, SQL, ETL/ELT pipelines, dimensional modeling, Tableau, Power BI

Data Engineer

Webhead Technologies — San Antonio, TX, USA

May 2011 – November 2015

- ▮ Developed and maintained foundational **ETL pipelines** using **Python** and **SQL** to process structured and semi-structured application data for analytics and internal reporting systems.
- ▮ Designed SQL-based reporting datasets supporting recurring operational dashboards and business intelligence needs.
- ▮ Applied advanced **SQL techniques** (joins, aggregations, indexing strategies, performance tuning) to support ad-hoc analytical requests and product usage analysis.
- ▮ Collaborated closely with application engineering teams to ensure data accuracy, schema consistency, and reliable integration between source systems and reporting layers.
- ▮ Assisted in early-stage dimensional modeling efforts to establish structured reporting schemas supporting scalable analytics growth.
- ▮ Supported data validation and quality assurance initiatives to improve reporting accuracy and maintain trust in business intelligence outputs.

Environment: Python, SQL, relational databases

EDUCATION

▮ Bachelor of Science in Computer Science

University of Central Florida — Orlando, FL, USA

August 2007 – May 2011